

1 Simulation study*

We compare the Bayesian fusion forest on the combined RCT and RWD data with a Bayesian causal forest on the trial alone. We quantify the reduction in CATE estimation error that the real-world data provide. We vary the strength of the unmeasured confounding in the real-world data.

1.1 Simulation setup

We draw $p = 10$ covariates $X \sim \mathcal{N}(0, \Sigma)$ in both sources. We impose an AR(1) correlation $\Sigma_{ij} = \rho^{|i-j|}$ with $\rho = 0.3$. We use five active covariates and five noise covariates. We generate the outcome:

$$\begin{aligned}\log T_{\text{RCT}} &= m_0(X) + A\tau(X) + \varepsilon_{\text{RCT}}, \\ \log T_{\text{RWD}} &= m_0(X) + \lambda_d d(X) + A\tau(X) + \lambda_u AU + \varepsilon_{\text{RWD}},\end{aligned}\tag{1}$$

with components:

$$m_0(X) = 4X_1 - 2X_2X_3 + X_4^2, \quad \tau(X) = 1/2 + 2X_1 - X_2^2, \quad d(X) = 2X_4 - X_5.\tag{2}$$

The function τ is the true CATE on the log-time scale. We randomise treatment in the trial, $A \sim \text{Bernoulli}(1/2)$. We let real-world treatment depend on an unmeasured confounder $U \sim \text{Uniform}(0, 1)$. We assign $A \sim \text{Bernoulli}(\text{expit}(X_1 + U))$. The confounder enters the outcome only in the treated arm. We control the between-study heterogeneity through λ_d . We control the confounding through λ_u . We model the trial error as $\varepsilon_{\text{RCT}} \sim \mathcal{N}(0, \sigma^2)$ with $\sigma = 3/4$. We draw the real-world error from a standardised Gumbel law. We set $\varepsilon_{\text{RWD}} = \beta \gamma_E - G$ with $G \sim \text{Gumbel}(0, \beta)$ and $\beta = \sqrt{6}/\pi$. This error has mean zero and unit variance. Trial and real-world event times are log-normal and Weibull, respectively. We right-censor the trial by an exponential censoring time, tuned to about 35% censoring. We interval-censor the real-world data at eight inspection times. Events past the last inspection are right-censored. This contributes to circa 20% of the RWD observations.

We fit the Bayesian fusion forest to the combined data with default parameters. The single-source baseline is an accelerated failure time Bayesian causal forest, fitted to the trial and to the real-world data separately (?). We run 1000 Monte Carlo replications. Each fit draws 5000 posterior samples after 5000 burn-in iterations. Our primary metric is the root mean squared error of the CATE. This is the precision in estimation of heterogeneous effects (PEHE) introduced by ?. We also report coverage, posterior variance and bias.

1.2 Simulation results

We first vary the confounding strength λ_u and hold the between-study heterogeneity fixed at $\lambda_d = 1$. Figure 1 reports the results. The fusion forest stays unbiased across all confounding levels. The real-world-only estimate becomes biased as the confounding strengthens. The trial-only estimate is unaffected by the level of confounding. The fusion forest attains a lower RMSE than both single-source estimates. Its coverage runs above the nominal level, while the single-source coverage runs below it. The posterior variance follows the same pattern, too large for the fusion forest and too small for the single-source fits.

We then vary the between-study heterogeneity λ_d and hold the confounding fixed at $\lambda_u = 1$. Figure 2 reports the results.

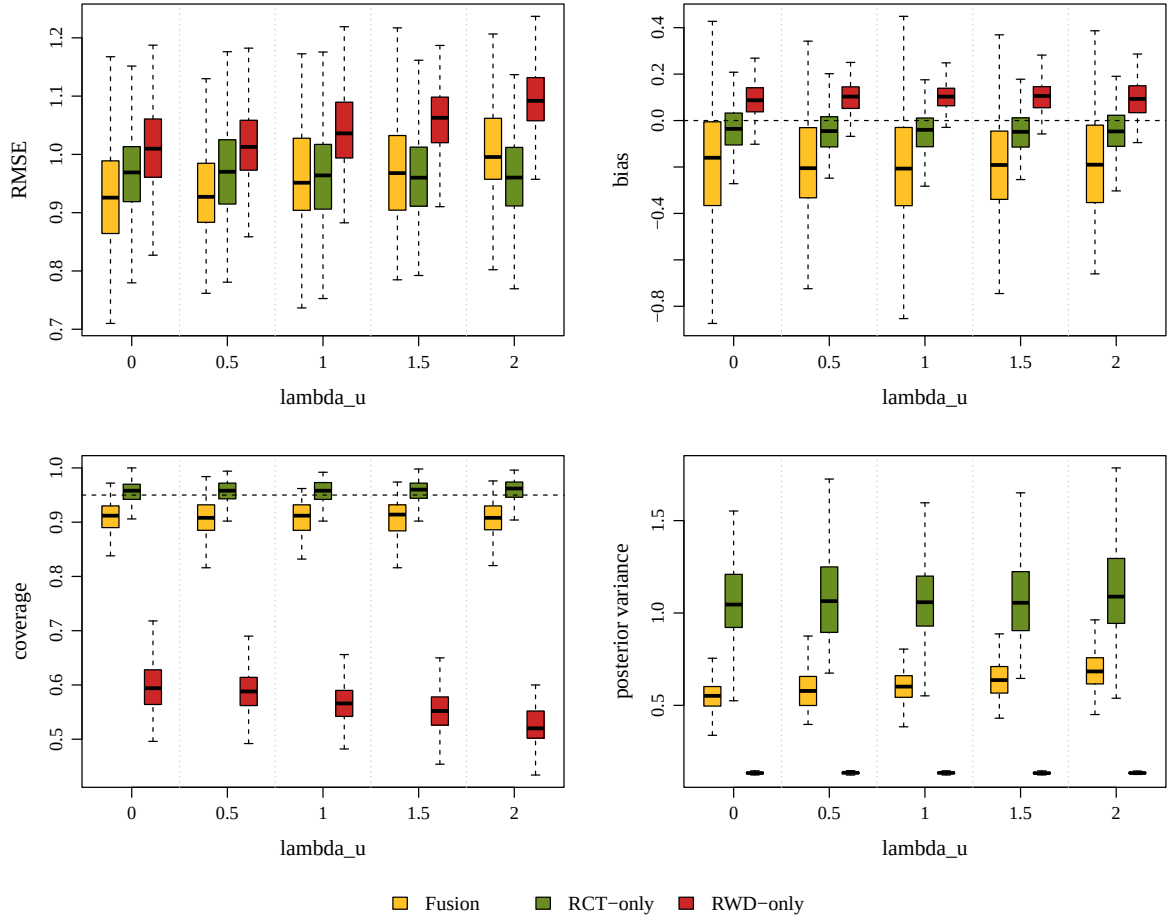


Figure 1: CATE metrics under varying confounding strength λ_u , with the between-study heterogeneity held at $\lambda_d = 1$. Each panel shows the RMSE, bias, coverage, or posterior variance for the Bayesian fusion forest and the two single-source baselines.

The fusion forest lowers the RMSE substantially in both studies. The fusion coverage is too high, and the single-source coverage too low. Crucially, the fusion estimate is robust to both the confounding and the between-study heterogeneity.

1.3 Sidenote

This is a first setup for the simulation study. It works reasonably well. The Bayesian fusion forest still shows a large posterior variance. The same issue appears as overcoverage of the credible intervals. (It is not due to nonparametric error distribution or the censoring. Probably prior calibration or DGP.) This is something to fix.

The list below collects possible extensions.

- Compare the nonparametric error model with a single Gaussian.
- Vary the sample size, or vary the RWD-to-RCT size ratio.
- Vary the complexity of the treatment-effect function, down to a homogeneous effect.
- Introduce covariate shift between the two sources.

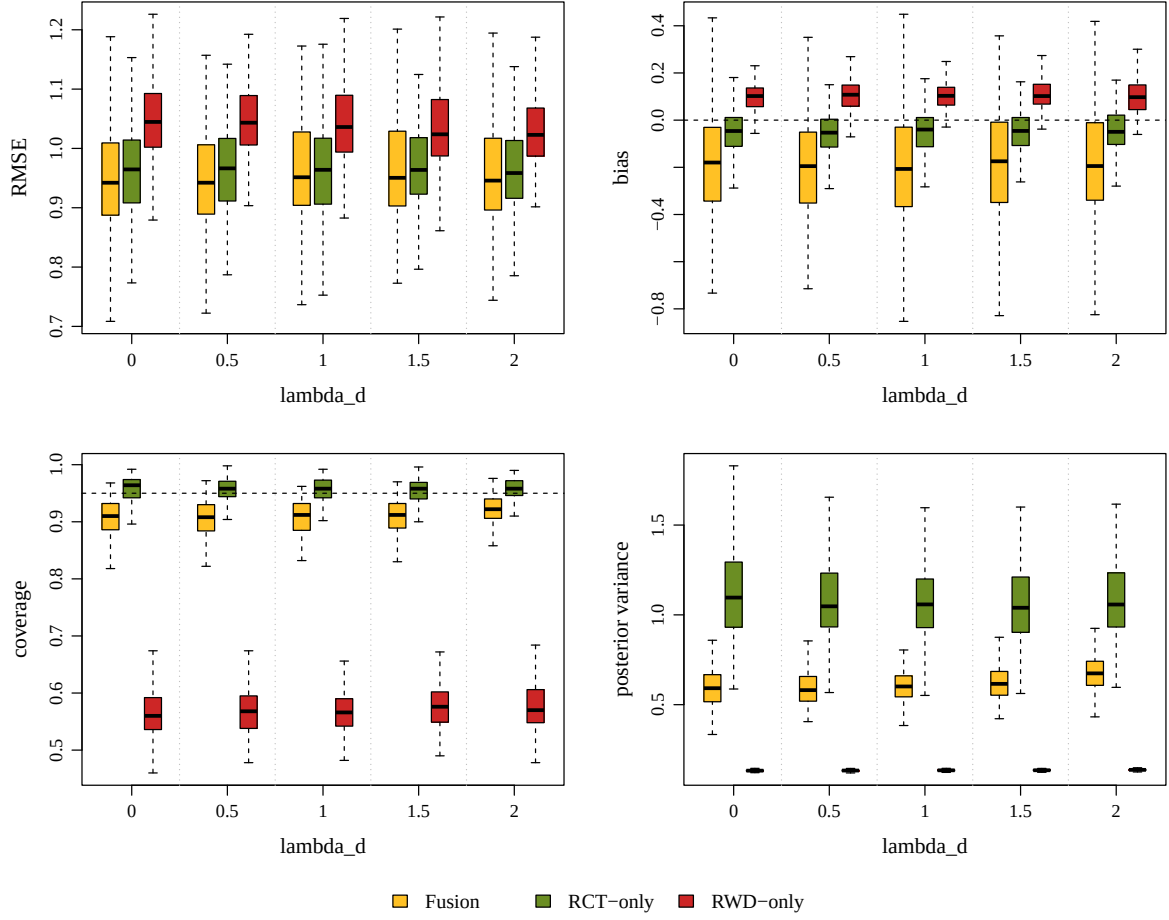


Figure 2: CATE metrics under varying between-study heterogeneity λ_d , with the confounding held at $\lambda_u = 1$. Panels and methods as in Figure 1.